

The Twin Prime Conjecture: Refutation of a Claimed Proof

Abstract

We examine the claimed proof of the Twin Prime Conjecture given in the accompanying document. The argument uses the circle method, Vaughan's identity, and a contraction fixed-point argument. We prove rigorously that the contraction argument is logically circular and that the proof does not establish the claimed asymptotic. Consequently, the Twin Prime Conjecture remains unproven.

1. Notation and Restatement

Let $\Lambda(n)$ be the von Mangoldt function. Define

$$R(x) = \sum_{n \leq x} \Lambda(n) \Lambda(n+2).$$

Let

$$\mathfrak{S}(2) = 2C_2 = 2 \prod_{p>2} (1 - (p-1)^{-2})$$

be the Hardy-Littlewood twin prime constant. The claim asserts

$$R(x) = \mathfrak{S}(2)x + o(x)$$

as

$$x \rightarrow \infty$$

, from which it follows that there are infinitely many twin primes.

The claimed proof defines the residual

$$\Delta(x) = R(x) - \mathfrak{S}(2)x,$$

an operator

$$\mathcal{T}[f](x) = \int_1^x f(t) \left(\sum_{n \leq x/t} \frac{\Lambda(n)}{n^2} \right) \frac{dt}{t},$$

and an auxiliary function

$$\mathcal{E}(x) = R(x) - \mathfrak{S}(2)x - \int_1^x \Delta(t) \left(\sum_{n \leq x/t} \frac{\Lambda(n)}{n^2} \right) \frac{dt}{t}.$$

It then claims that

$$\mathcal{T}$$

is a contraction on the Banach space

$$\mathcal{X} = \{f : [2, \infty) \rightarrow \mathbb{R} : \|f\|_{\mathcal{X}} := \sup_{x \geq 2} |f(x)|(\log x)^2/x < \infty\},$$

that

$$\mathcal{E}(x) \ll x/(\log x)^2$$

, and hence via the Neumann series that

$$\Delta(x) \ll x/(\log x)^2.$$

We shall prove that this argument is logically circular.

2. The Definition of

$$\mathcal{E}$$

Is Self-Referential

Recall the definition of

$$\mathcal{E}$$

:

$$\mathcal{E}(x) = \Delta(x) - \mathcal{T}[\Delta](x).$$

Substituting the definition of

$$\Delta(x)$$

gives

$$\mathcal{E}(x) = R(x) - \mathfrak{S}(2)x - \int_1^x (R(t) - \mathfrak{S}(2)t) \left(\sum_{n \leq x/t} \frac{\Lambda(n)}{n^2} \right) \frac{dt}{t}.$$

Now consider the structure of this expression. The quantity

$$R(x)$$

is the object whose asymptotic we are trying to determine. The claimed proof asserts (Lemma 11.1) that

$$\mathcal{E}(x) \ll x/(\log x)^2$$

without having established the asymptotic for

$$R(x).$$

Lemma 2.1. The claimed bound

$$\mathcal{E}(x) \ll x/(\log x)^2$$

is not established independently of the asymptotic

$$R(x) \sim \mathfrak{S}(2)x.$$

Proof. The minor arc contribution to

$$\mathcal{E}$$

is controlled by the large-sieve bound, giving

$$O(x \exp(-c\sqrt{\log x}))$$

, which is indeed admissible. However, the major arc contribution to

$$\mathcal{E}$$

involves the interaction of the oscillatory integrals

$$I(\beta) = \sum_{n \leq x} e(n\beta) = \int_0^x e(t\beta) dt + O(1)$$

with the character sums

$$\sum_{\substack{\chi \bmod q \\ \chi \neq \chi_0}} \frac{\overline{\chi}(-2)}{\tau(\overline{\chi})} \sum_{n \leq x} \chi(n) \Lambda(n) e(n\beta).$$

The standard circle method extracts the main term

$$\mathfrak{S}(2)x$$

from the principal characters and bounds the non-principal contribution by Siegel–Walfisz. The convolution integral

$$\int \Delta(t) \cdots dt/t$$

does not appear in the standard analysis; it is constructed by separating a purported “cross-character” interaction term and identifying it with

$$\mathcal{T}[\Delta]$$

For this identification to be valid, one must interchange the order of summation over characters, integration over

$$\beta$$

, and the summation implicit in

$$\Delta(t)$$

. Such an interchange is justified only if one already knows sufficient decay of

$$\Delta(t)$$

— precisely the information the argument seeks to establish.

Formally, one would need to prove

$$\begin{aligned} \sum_{\chi \neq \chi_0} \frac{\bar{\chi}(-2)}{\tau(\bar{\chi})} \int_{-Q/x}^{Q/x} \sum_{n \leq x} \chi(n) \Lambda(n) e(n\beta) \sum_{m \leq x} \bar{\chi}(m) \Lambda(m) e(-m\beta) e(-2a/q) d\beta \\ = \int_1^x \Delta(t) \left(\sum_{n \leq x/t} \frac{\Lambda(n)}{n^2} \right) \frac{dt}{t} + \text{admissible error.} \end{aligned}$$

No such proof is given. The equality is asserted, not derived.

□

3. The Contraction Claim Is Not Proved

The claimed proof asserts (Lemma 7.1) that $\|\mathcal{T}\|_{x \rightarrow x} \leq C/\log 2 < 1$ and further (Lemma 11.2) that $\|\mathcal{T}[f]\|_x \leq \frac{1}{2}\|f\|_x$ for all large x .

Lemma 3.1. Under the definitions given, the estimate $\|\mathcal{T}[f]\|_x \leq \frac{1}{2}\|f\|_x$ is not validly derived from the Prime Number Theorem alone.

Proof. The calculation in the claimed proof proceeds as follows:

$$\frac{|\mathcal{T}[f](x)|(\log x)^2}{x} \leq \frac{(\log x)^2}{x} \int_1^x \frac{|f(t)|}{t} \left(\sum_{n \leq x/t} \frac{\Lambda(n)}{n^2} \right) dt.$$

Using $|f(t)| \leq \|f\|_x \cdot t/(\log t)^2$ and $\sum_{n \leq y} \Lambda(n)/n^2 = 1 + O(1/\log y)$, we obtain

$$\frac{|\mathcal{T}[f](x)|(\log x)^2}{x} \leq \|f\|_x \cdot \frac{(\log x)^2}{x} \int_1^x \frac{dt}{(\log t)^2} + \mathcal{R}(x),$$

where $\mathcal{R}(x)$ accounts for the error $O(1/\log y)$ in the PNT estimate.

Now

$$\int_1^x \frac{dt}{(\log t)^2} = \frac{x}{(\log x)^2} + \frac{2x}{(\log x)^3} + O\left(\frac{x}{(\log x)^4}\right).$$

This is a standard asymptotic obtained by integration by parts. Substituting,

$$\frac{(\log x)^2}{x} \int_1^x \frac{dt}{(\log t)^2} = 1 + \frac{2}{\log x} + O\left(\frac{1}{(\log x)^2}\right).$$

Therefore

$$\frac{|\mathcal{T}[f](x)|(\log x)^2}{x} \leq \|f\|_x \left(1 + \frac{2}{\log x} + O\left(\frac{1}{(\log x)^2}\right)\right) + |\mathcal{R}(x)|.$$

The leading constant is 1, not $\frac{1}{2}$ as claimed. To obtain a contraction factor strictly less than 1, one would need a nontrivial estimate that reduces the coefficient below unity — for instance, by exploiting cancellation in the convolution kernel, or by proving a stronger bound on the PNT remainder. No such estimate is provided.

The error term $\mathcal{R}(x)$ coming from the PNT remainder $O(1/\log y)$ contributes at most $O(1/\log x)$ multiplicatively, which is insufficient to bring the factor below 1.

Thus the claimed contraction is not proved; the operator $\mathcal{T}$ is shown to be at best non-expansive: $\|\mathcal{T}[f]\|_x \leq \|f\|_x(1 + o(1))$. $\square$

Corollary 3.2. The Neumann series $\Delta = \sum_{k=0}^{\infty} \mathcal{T}^k[\mathcal{E}]$ does not converge in $\mathcal{X}$ under the estimates provided.

Proof. For the Neumann series to converge, one needs $\lim_{k \rightarrow \infty} \|\mathcal{T}^k[\mathcal{E}]\|_x = 0$. With $\|\mathcal{T}\| \approx 1$, the terms do not decay geometrically and the series may diverge. Even with $\|\mathcal{T}\| < 1$ unproved, the convergence is not established. $\square$

4. The Circularity Is Fatal: Proof That the Argument Does Not Establish $R(x) \sim \mathfrak{S}(2)x$

We now give a formal proof that the contraction argument is circular and cannot establish the claimed asymptotic.

Theorem 4.1. Assume only the estimates from the circle method (Sections 1-5 of the claimed proof) and the Prime Number Theorem. Then the fixed-point equation $\Delta = \mathcal{T}[\Delta] + \mathcal{E}$ together with the estimate $\mathcal{E}(x) \ll x/(\log x)^2$ does **not** logically imply $\Delta(x) = o(x)$ without additional assumptions that are themselves equivalent to or stronger than the desired conclusion.

Proof. We proceed by constructing an explicit counterexample to the logical implication. That is, we exhibit a function $\tilde{R}(x)$ that satisfies all the estimates established in Sections 1-5 of the claimed proof, satisfies the fixed-point equation with the same formal structure, yet does **not** satisfy $\tilde{R}(x) \sim \mathfrak{S}(2)x$.

Let $\delta > 0$ be a small fixed constant. Define

$$\tilde{\Delta}(x) = \delta x,$$

and set

$$\tilde{R}(x) = \mathfrak{S}(2)x + \tilde{\Delta}(x).$$

Define $\tilde{\mathcal{E}}$ by the fixed-point equation:

$$\tilde{\mathcal{E}}(x) = \tilde{\Delta}(x) - \mathcal{T}[\tilde{\Delta}](x).$$

We compute $\mathcal{T}[\tilde{\Delta}]$:

$$\mathcal{T}[\tilde{\Delta}](x) = \int_1^x \delta t \left(\sum_{n \leq x/t} \frac{\Lambda(n)}{n^2} \right) \frac{dt}{t} = \delta \int_1^x \left(\sum_{n \leq x/t} \frac{\Lambda(n)}{n^2} \right) dt.$$

Let $F(y) = \sum_{n \leq y} \Lambda(n)/n^2$. From the PNT we have $F(y) = 1 + O(1/\log y)$. Then

$$\mathcal{T}[\tilde{\Delta}](x) = \delta \int_1^x (1 + O(1/\log t)) dt = \delta x + \delta \cdot O\left(\frac{x}{\log x}\right).$$

Thus

$$\tilde{\mathcal{E}}(x) = \delta x - \delta x + \delta \cdot O\left(\frac{x}{\log x}\right) = O\left(\frac{x}{\log x}\right).$$

Now observe:

- (1) $\tilde{\mathcal{E}}(x) \ll x/\log x$ — this is **compatible** with the claimed bound $\mathcal{E}(x) \ll x/(\log x)^2$ (the $x/\log x$ bound is actually weaker, so the claimed stronger bound does not exclude this scenario).

- (2) The function $\widetilde{\Delta}(x) = \delta x$ satisfies $\|\widetilde{\Delta}\|_{\mathcal{X}} = \delta(\log x)^2 \rightarrow \infty$, so $\widetilde{\Delta} \notin \mathcal{X}$ — but the question is whether the fixed-point equation **forces** $\Delta \in \mathcal{X}$ with small norm, not whether $\widetilde{\Delta}$ itself is in $\mathcal{X}$.
- (3) More importantly, consider the function $\widetilde{\Delta}_{\varepsilon}(x) = \varepsilon \cdot x/(\log x)$ for any fixed $\varepsilon > 0$. Then $\|\widetilde{\Delta}_{\varepsilon}\|_{\mathcal{X}} = \varepsilon \log x \rightarrow \infty$, so this too is not in $\mathcal{X}$. But the crucial observation is that both $\widetilde{\Delta}(x) = \delta x$ and $\widetilde{\Delta}_{\varepsilon}(x) = \varepsilon x/\log x$ satisfy

$$\mathcal{T}[\widetilde{\Delta}](x) = \widetilde{\Delta}(x) + O\left(\frac{x}{(\log x)^2}\right),$$

meaning the fixed-point equation $\Delta = \mathcal{T}[\Delta] + \mathcal{E}$ with $\mathcal{E}(x) \ll x/(\log x)^2$ is satisfied by **any** Δ that grows linearly (or slightly sublinearly) in x , because $\mathcal{T}$ acts approximately as the identity on such functions.

Specifically, for any function $g(x)$ satisfying $|g(x)| \ll x$ and sufficient regularity, we have

$$\mathcal{T}[g](x) = g(x) + O\left(\frac{x}{\log x}\right)$$

by the same PNT calculation. The operator $\mathcal{T}$ is essentially the identity at leading order, so the equation $\Delta = \mathcal{T}[\Delta] + \mathcal{E}$ becomes

$$\Delta = \Delta + O\left(\frac{x}{\log x}\right) + \mathcal{E}(x),$$

which yields $\mathcal{E}(x) = O(x/\log x)$ regardless of the actual asymptotic behavior of $\Delta(x)$. The fixed-point equation places **no constraint** on the leading asymptotic of $\Delta(x)$; it merely relates the subleading corrections.

Conclusion of Theorem 4.1. The fixed-point equation cannot determine whether $\Delta(x) = o(x)$ or $\Delta(x) \sim \delta x$, because the operator $\mathcal{T}$ is asymptotic to the identity on functions of linear growth. The argument therefore does **not** prove $\Delta(x) = o(x)$. $\square$

5. The Parity Barrier: Why the Bilinear Method Cannot Reach Gap 2

We now provide a rigorous explanation of why the bilinear estimates in the claimed proof cannot yield the twin prime asymptotic, even if the contraction argument were valid.

Theorem 5.1 (Parity Barrier, quantitative form). Let $A(n)$ be the indicator of the condition “ $\omega(n) \equiv 0 \pmod{2}$ ” where $\omega(n)$ counts distinct prime factors. Then for any bilinear form

$$B(\alpha) = \sum_{m \sim M} \sum_{n \sim N} \xi_m \eta_n e(mn\alpha)$$

with $|\xi_m|, |\eta_n| \ll_\varepsilon x^\varepsilon$ and $MN = x$, the large-sieve bound

$$|B(\alpha)| \ll x \exp(-c\sqrt{\log x})$$

for $\alpha \in \mathfrak{m}$ and the corresponding major-arc estimate together yield

$$\sum_{n \leq x} A(n) \Lambda(n) \Lambda(n+2) = \frac{1}{2} \mathfrak{S}(2)x + o(x),$$

and similarly

$$\sum_{n \leq x} (1 - A(n)) \Lambda(n) \Lambda(n+2) = \frac{1}{2} \mathfrak{S}(2)x + o(x).$$

That is, the method gives the **same** asymptotic for numbers with an even number of prime factors as for those with an odd number. It cannot distinguish the case $\omega(n) = 1$ (primes) from $\omega(n) = 3, 5, 7, \dots$ (products of three or more primes).

Proof. This is the classic parity phenomenon discovered by Selberg. The bilinear form depends only on the coefficients ξ_m, η_n . The large-sieve bound depends on the ℓ^2 norm of these coefficients — $(\sum |\xi_m|^2)^{1/2} (\sum |\eta_n|^2)^{1/2}$ — which is unchanged under replacing Λ by $\Lambda \cdot A$ or $\Lambda \cdot (1 - A)$, since both have the same ℓ^2 norm up to a factor $1/2 + o(1)$.

More precisely, the bilinear large-sieve inequality gives

$$\int_{\mathfrak{m}} |B(\alpha)|^2 d\alpha \ll x (\log x)^{-A}$$

and the major-arc analysis gives a singular series that depends on the mean values $\sum \xi_m/m$ and $\sum \eta_n/n$. Changing Λ to $\Lambda \cdot A$ replaces these mean values by (approximately) one-half of the original, leading to the factor $1/2$ in the asymptotic.

For the twin prime problem, we need to isolate n with $\omega(n) = 1$ and $\omega(n+2) = 1$, not merely $A(n) = A(n+2) = 1$ (both even) or both odd. The parity barrier says that Type I/II methods give the same leading term for both parity classes, so they cannot take the final step from “odd number of prime factors” to “exactly one prime factor.” $\square$

Theorem 5.2. The bilinear estimate in the claimed proof (Lemma 4.1) is of Type I/II and therefore subject to the parity barrier. It does not provide the

parity discrimination needed to isolate twin primes from other configurations with the same parity.

Proof. The term $S_2(\alpha)$ is

$$S_2(\alpha) = \sum_{U < m \leq x/U} \sum_{U < n \leq x/m} \xi_m \eta_n e(mn\alpha)$$

with $\xi_m = \Lambda(m)$ and $\eta_n = \sum_{d|n, d \leq U} \mu(d)$. This is a bilinear form in the variables m and n . The bound in Lemma 4.1 uses the large sieve, which gives an estimate depending on $\|\xi\|_2 \|\eta\|_2$. No parity information enters beyond the ℓ^2 norm.

The coefficients $\eta_n = \sum_{d|n, d \leq U} \mu(d)$ are a truncated Möbius sum, which is closely related to the indicator of numbers with no small prime factors (the Buchstab sieve weight). This coefficient is used to sift out numbers with small prime factors, but it does not count prime factors modulo 2 — it treats all small primes equally. The parity of $\omega(n)$ is invisible to this coefficient beyond the squarefree-sieve level. $\square$

6. The Level-of-Distribution Gap

We now quantify the gap between what the claimed proof establishes and what would be required.

Definition 6.1 (Level of Distribution). An arithmetic function $a(n)$ is said to have level of distribution θ if for every $A > 0$,

$$\sum_{q \leq x^\theta} \max_{(a,q)=1} \left| \sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} a(n) - \frac{1}{\phi(q)} \sum_{n \leq x} a(n) \right| \ll_A x (\log x)^{-A}.$$

The Bombieri-Vinogradov theorem states that the primes (with weight $\Lambda(n)$) have level of distribution $\theta = 1/2 - \varepsilon$. The Elliott-Halberstam conjecture asserts $\theta = 1 - \varepsilon$.

Theorem 6.1. To extract the Hardy-Littlewood singular series $\mathfrak{S}(2)x$ as a **lower bound** (rather than merely an upper bound) for the twin prime correlation $R(x)$ using the circle method, one needs a level of distribution $\theta > 1/2$ for the primes.

Proof. In the circle method, the major arcs are defined by $q \leq Q$. After evaluating the major arc contribution, the main term is

$$\mathfrak{S}(2, Q) \cdot x, \quad \text{where} \quad \mathfrak{S}(2, Q) = \sum_{\substack{q \leq Q \\ (q, 2) = 1}} \frac{\mu(q)}{\phi(q)^2}.$$

The truncation error from the tail $q > Q$ is

$$\sum_{\substack{q > Q \\ (q, 2) = 1}} \frac{|\mu(q)|}{\phi(q)^2} \ll \frac{1}{Q^{1-\varepsilon}}.$$

Thus the truncated singular series satisfies

$$\mathfrak{S}(2, Q) = \mathfrak{S}(2) + O(Q^{-1+\varepsilon}).$$

If one only controls the major arcs for $q \leq (\log x)^A$ (as Siegel-Walfisz gives), then $Q = (\log x)^A$ and the truncation error is $O((\log x)^{-A(1-\varepsilon)})$, which is merely $o(x)$ — but then the total major arc integration error from the non-principal characters also only has log-power savings, and the full major arc contribution is

$$\mathfrak{S}(2)x + O\left(\frac{x}{(\log x)^{A/2}}\right),$$

which is **compatible** with $R(x) = \mathfrak{S}(2)x + \delta x$ for any $\delta > 0$, because the error $O(x/(\log x)^{A/2})$ cannot exclude a linear-order deviation.

To exclude a linear-order deviation, one needs that the major arcs cover enough moduli q so that both the singular series truncation error and the character-sum errors are $o(x)$. This requires $Q \rightarrow \infty$ as a positive power of x , i.e., $Q = x^\theta$ with $\theta > 0$. But then the character sum

$$\sum_{q \leq x^\theta} \sum_{\substack{\chi \bmod q \\ \chi \neq \chi_0}} \left| \sum_{n \leq x} \chi(n) \Lambda(n) \right|$$

must be bounded by $o(x)$. This is precisely the level-of-distribution problem.

For the twin prime problem in particular, the lower bound requires controlling the contribution of the **singular series tail** to the extent that it doesn't mask the positive constant $\mathfrak{S}(2)$. This requires $\theta > 1/2$. The claimed proof does not establish this.

Theorem 6.2. The major arc analysis in the claimed proof uses Siegel-Walfisz (valid for $q \leq (\log x)^A$) and does not establish a level of distribution $\theta > 0$ for the twin prime correlation $R(x)$.

Proof. The major arc contribution is computed as

$$\sum_{q \leq Q} \frac{\mu(q)^2}{\phi(q)^2} c_q(-2) \cdot x + \text{error},$$

where $c_q(-2)$ is the Ramanujan sum. The error term is controlled by

$$\sum_{q \leq Q} \frac{1}{\phi(q)} \max_{(a,q)=1} \left| \psi(x; q, a) - \frac{x}{\phi(q)} \right|.$$

Siegel-Walfisz bounds each term by $x \exp(-c\sqrt{\log x})$ for $q \leq (\log x)^A$, giving an error $O(Qx \exp(-c\sqrt{\log x}))$. With $Q = (\log x)^A$, this is $o(x)$. But the singular series truncation error with $Q = (\log x)^A$ is $O(x/(\log x)^A)$, not $o(x/(\log x)^2)$.

The claimed proof asserts $Q = x^\theta$ with $\theta > 1/2$ (page 2). But the subsequent analysis (page 4-5) only invokes Siegel-Walfisz, which is valid for $q \leq (\log x)^A$, not for $q \leq x^\theta$. The extension from $(\log x)^A$ to x^θ is not justified.

One might argue that Bombieri-Vinogradov extends the range to $x^{1/2}/(\log x)^B$. But Bombieri-Vinogradov applies to averages over **all** a coprime to q , not to the specific shifted correlation $n+2$. To use Bombieri-Vinogradov for the twin prime problem, one needs to handle the shift $h = 2$, which introduces an additional technical complication: one must control

$$\sum_{q \leq Q} \frac{1}{\phi(q)} \max_{(a,q)=1} \left| \psi_2(x; q, a) - \frac{x}{\phi(q)} \right|,$$

where $\psi_2(x; q, a) = \sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \Lambda(n) \Lambda(n+2)$.

This is **not** the standard Bombieri-Vinogradov setting. The standard result controls $\Lambda(n)$ alone, not the correlation $\Lambda(n) \Lambda(n+2)$. Extending Bombieri-Vinogradov to correlations is a major open problem. The claimed proof does not address this extension. $\square$

7. The Analytic Continuation Is Not Proved

The claimed Lemma 6.1 asserts that the Dirichlet series

$$F(s) = \sum_{n=1}^{\infty} \frac{\Lambda(n) \Lambda(n+2)}{n^s}$$

admits meromorphic continuation to $\Re(s) > 1/2$ with a simple pole at $s = 1$ and no other poles in $\Re(s) \geq 1$.

Theorem 7.1. The meromorphic continuation claimed in Lemma 6.1 is not established by the arguments in the claimed proof.

Proof. The Dirichlet series $F(s)$ is

$$F(s) = \sum_{n=1}^{\infty} \frac{\Lambda(n)\Lambda(n+2)}{n^s} = \sum_{n=1}^{\infty} \frac{\Lambda(n-2)\Lambda(n)}{(n-2)^s} \mathbf{1}_{n>2}.$$

For $\Re(s) > 1$, the absolute convergence allows rearrangement. Using Vaughan's identity, one decomposes Λ into $a_1 + a_2 - a_3$ and obtains a multilinear expression. Each term involves sums of the form

$$\sum_m \frac{\xi_m}{m^s} \sum_{n:mn+2=r} \frac{\eta_n}{n^s},$$

which is not naturally a product of L -functions, but rather a convolution shifted by 2.

To convert this into L -functions, the claimed proof inserts additive characters:

$$\sum_m \frac{\xi_m}{m^s} \sum_n \frac{\eta_n}{(mn+2)^s} = \int_0^1 \sum_m \frac{\xi_m e(m\alpha)}{m^s} \sum_n \frac{\eta_n e(n\alpha)}{n^s} e(2\alpha) d\alpha.$$

But the separation into m^s and n^s in the denominator is **not** valid; the denominator is $(mn+2)^s$, not $m^s n^s$. Therefore

$$\frac{\xi_m \eta_n}{(mn+2)^s} \neq \frac{\xi_m}{m^s} \cdot \frac{\eta_n}{n^s}$$

except in approximate form for large mn , and the approximation error must be controlled.

One can write

$$\frac{1}{(mn+2)^s} = \frac{1}{(mn)^s} \left(1 + \frac{2}{mn}\right)^{-s} = \frac{1}{(mn)^s} \sum_{k=0}^{\infty} \binom{-s}{k} \left(\frac{2}{mn}\right)^k,$$

expanding in powers of $2/(mn)$. Then

$$F(s) = \sum_{k=0}^{\infty} \binom{-s}{k} 2^k \sum_{m,n} \frac{\Lambda(m)\Lambda(n)}{(mn)^{s+k}}.$$

The inner sum factors as $(-\zeta'(s+k)/\zeta(s+k))^2$, leading to an expression involving derivatives of the Riemann zeta function. However, the k -sum converges only

for $\Re(s) > 0$ with careful estimation, and the analytic properties of the resulting series are not standard. The claimed continuation to $\Re(s) > 1/2$ with only a pole at $s = 1$ requires information about the distribution of the zeros of $\zeta(s)$ (or the associated Dirichlet polynomials) that is not available unconditionally.

The claimed derivation using L -functions through the circle-method orthogonality is also incompletely justified, since the circle method applies to finite sums $n \leq x$, not to the full Dirichlet series. The passage from finite exponential sums to the infinite L -function requires a Perron-type limiting argument that commutes the limit $x \rightarrow \infty$ with the character sum, which is delicate and not provided.

Thus Lemma 6.1 is not proved. $\square$

8. The Quantitative Lower Bound Is Unjustified

Theorem 8.1. The claimed bound

$$T(x) \geq \frac{2C_2x}{(\log x)^2} \left(1 + O\left(\frac{1}{\log \log x} \right) \right)$$

does not follow from the preceding arguments, even if those arguments were valid.

Proof. The step from the correlation $R(x)$ to the counting function $T(x)$ uses the inequality

$$\sum_{\substack{p \leq x \\ p, p+2 \text{ prime}}} (\log p)(\log(p+2)) \leq T(x)(\log x)^2.$$

This is correct. Combined with $R(x) \sim \mathfrak{S}(2)x = 2C_2x$, one obtains

$$T(x) \geq \frac{2C_2x}{(\log x)^2} (1 + o(1)).$$

However, the error term $o(1)$ in the claimed proof is specified as $O(1/\log \log x)$, which is a **quantitative** claim. To obtain this quantitative error, one needs explicit control on the difference between the weighted and unweighted counts.

The standard identity is

$$\sum_{\substack{p \leq x \\ p, p+2 \text{ prime}}} (\log p)(\log(p+2)) = (\log x)^2 T(x) - 2 \int_2^x \frac{\log t}{t} T(t) dt + \text{error}.$$

Let $T(x) \sim cx/(\log x)^2$ for some $c > 0$ (this is what we are trying to prove). Then

$$\int_2^x \frac{\log t}{t} T(t) dt \sim c \int_2^x \frac{dt}{t \log t} = c \log \log x,$$

which grows without bound. The weighted sum minus $(\log x)^2 T(x)$ is thus asymptotic to $-2cx \log \log x / (\log x)^2$, which is **not** negligible compared to the main term cx .

In other words, if one only knows the weighted sum is $\sim \mathfrak{S}(2)x$, then the unweighted count $T(x)$ could be as small as $x/(\log x)^{2+\delta}$ for any $\delta > 0$, and the weighted sum would still be $\sim \mathfrak{S}(2)x$ because most of the weighted contribution comes from n near x . The lower bound $x/(\log x)^2$ is the **maximal** possible order consistent with the weighted asymptotic, not a proven lower bound.

To prove $T(x) \gg x/(\log x)^2$, one needs additional regularity of T — specifically, one needs to know that T is approximately monotone or satisfies a differential inequality. No such regularity is established. $\square$

9. Summary of Errors

We have identified the following logical errors in the claimed proof:

Error 1: The fixed-point equation is underdetermined.

The operator T is asymptotic to the identity on the space of functions of linear growth. The fixed-point equation therefore reduces to $E(x) = O(x/\log x)$ regardless of the leading asymptotic of $\Delta(x)$. The equation does NOT force $\Delta(x) = o(x)$. (Theorem 4.1)

Error 2: The contraction norm estimate is incorrect.

The claimed contraction factor $1/2$ is not derived; the PNT-based estimate gives a norm of $1 + o(1)$, which is non-expansive at best. (Lemma 3.1)

Error 3: The bound on E is not independently established.

The definition of E involves Δ itself, and the claimed bound $E(x) \ll x/(\log x)^2$ is not established without presupposing control over Δ . (Lemma 2.1)

Error 4: The parity barrier is not addressed.

The bilinear estimates are of Type I/II and cannot isolate primes from other numbers with an odd number of prime factors. Crossing the parity barrier requires structure beyond that provided. (Theorems 5.1, 5.2)

Error 5: No level of distribution is established.

The major arc analysis uses Siegel-Walfisz (moduli up to $(\log x)^A$), but the twin prime problem requires a level of distribution $\theta > 1/2$. The claimed extension to x^θ is not justified. (Theorems 6.1, 6.2)

Error 6: The analytic continuation is not proved.

The Dirichlet series $F(s)$ is not shown to continue meromorphically to $\operatorname{Re}(s) > 1/2$. (Theorem 7.1)

Error 7: The quantitative bound does not follow.

Even if the correlation asymptotic were established, the lower bound $T(x) \geq 2C_2 x / (\log x)^2$ does not follow without additional regularity of $T(x)$. (Theorem 8.1)

10. Conclusion

The argument presented in the claimed proof of the Twin Prime Conjecture contains a central logical error: its contraction fixed-point argument is circular. The operator T is shown to be asymptotic to the identity, so the fixed-point equation cannot determine the leading asymptotic of the correlation function. This error cannot be repaired by refining the estimates; it is a structural flaw in the logical architecture of the argument.

Additional errors — concerning the parity barrier, the level of distribution, the analytic continuation, and the quantitative lower bound — provide independent reasons why the claimed proof is invalid.

The Twin Prime Conjecture remains an open problem in mathematics.

Appendix: Summary of Rigorous Estimates Used

For clarity, we collect the standard estimates that the first three blocks of the claimed proof correctly employ:

1. Hardy-Littlewood circle method: $R(x) = \int |S(\alpha)|^2 e(-2\alpha) d\alpha$.
2. Vaughans identity: $\Lambda(n) = a_1 + a_2 - a_3$ for $n > U$.
3. Large sieve bilinear bound: $|S_2(\alpha)| \ll x \exp(-c \sqrt{\log x / \log \log x})$.
4. Siegel-Walfisz theorem: $\sum \chi(n) \Lambda(n) \ll x \exp(-c \sqrt{\log x})$.

5. Prime Number Theorem: $\sum \Lambda(n)/n^2 = 1 + O(1/\log y)$.

These estimates are correct. The error in the claimed proof is the claim that they logically imply $R(x) \sim S(2)x$ via the contraction argument. They do not.
